

Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir. Apply the specified permissions using chmod and record the observations.

Permission	Regular File-Test	Observation	Directory -Test	Observation
400 (r--)	Try to read and modify the file	Can read; cannot modify.	Try to list and enter the directory	Cannot list or enter because x permission is absent.
200 (-w-)	Try to read and modify the file	Cannot read normally; can modify/write.	Try to list, enter, and create a file	Cannot list or enter; x permission is absent, so creation is not practically possible.
100 (--x)	Try to read and execute the file	Cannot read; execution is possible only if it is a suitable executable/script.	Try to list and enter the directory	Cannot list; can enter/traverse the directory.
500 (r-x)	Try to read and modify the file	Can read; cannot modify. Can execute if suitable.	Try to list and enter the directory	Can list and enter; cannot create/delete because w is absent.
600 (rw-)	Try to read and modify the file	Can read and modify; cannot execute.	Try to list, enter, and create a file	Cannot list or enter because x permission is absent; cannot practically create files.
700 (rwx)	Try to read, modify, and execute the file	Can read, modify, and execute if the file is executable.	Try to list, enter, create, and delete files	Can list, enter, create, modify, and delete files.

Questions & Answers

1. What does read (r) permission allow you to do with a regular file?

Ans: It allows you to view or read the contents of the file.

2. What does read (r) permission allow you to do with a directory?

Ans: It allows you to list the names of files and directories inside the directory.

3. What does write (w) permission mean for a regular file?

Ans: It allows you to modify or change the contents of the file.

4. What does write (w) permission mean for a directory?

Ans: It allows you to create, delete, and rename entries inside the directory, generally when combined with x permission.

5. What does execute (x) permission mean for a regular file?

Ans: It allows the file to be executed as a program or script, provided it is a suitable executable/script.

6. Why is execute (x) permission important for accessing a directory?

Ans: It provides traverse/search permission, allowing you to enter the directory and access items within it.

7. Which permission combination allows the owner to fully access a regular file?

Ans: 700 (rwx) - read, write, and execute.

8. Which permission combination allows the owner to fully access a directory?

Ans: 700 (rwx) - list, enter/traverse, create, delete, and manage entries.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory?

Ans: A regular file contains data, whereas a directory contains references/entries to files and other directories. Therefore, r, w, and x control different operations: for a file they control reading, modifying, and executing its contents; for a directory they control listing entries, modifying directory entries, and traversing/accessing the directory.